

Smyth's Theorem, the Plastic Number, and Pandrosion Mahler Minimality

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Abstract

Smyth's theorem (1971) identifies $M(x^3 - x - 1) = \theta_0 \approx 1.32472$ (the plastic number) as the minimum Mahler measure over all non-reciprocal integer polynomials. We verify this via the Pandrosion–Jensen formula (Paper 59), exhaustive search over small-degree polynomials, and place it in the Mahler hierarchy: Kronecker $M = 1$ (Paper 80) < Lehmer ≈ 1.176 (reciprocal, Paper 44) < Smyth ≈ 1.325 (non-reciprocal). The Pandrosion field F_P of Smyth's polynomial has a unique real root $\theta_0 > 1$ with two complex conjugates inside S^1 , making θ_0 simultaneously the smallest Pisot number of degree 3 and a Pandrosion fixed point.

1 Smyth's theorem

Theorem 1.1 (Smyth, 1971). *Among all monic irreducible non-reciprocal polynomials $P \in \mathbb{Z}[x]$:*

$$\min M(P) = M(x^3 - x - 1) = \theta_0 \approx 1.32471795724 \dots \quad (1)$$

where θ_0 is the unique real root of $x^3 - x - 1$ (the plastic number).

Definition 1.2. A polynomial is *reciprocal* if $P(z) = \pm z^n P(1/z)$, i.e., its coefficient sequence is palindromic (up to sign). $P = x^3 - x - 1$ has coefficients $[1, 0, -1, -1]$ vs reversed $[-1, -1, 0, 1]$: non-reciprocal.

2 The plastic number

The plastic number θ_0 satisfies:

$$\theta_0^3 = \theta_0 + 1, \quad \theta_0 = 1.32471795724 \dots \quad (2)$$

Verified: $\theta_0^3 - \theta_0 - 1 = \mathcal{O}(10^{-16})$. The two complex conjugate roots have modulus $\approx 0.869 < 1$, so $M(x^3 - x - 1) = \theta_0$. This makes θ_0 the smallest *Pisot number* of degree 3: an algebraic integer > 1 whose conjugates are all inside the unit disk.

3 The Mahler hierarchy

Theorem	M	Type	Paper
Kronecker	$= 1$	All cyclotomic (reciprocal)	80
Lehmer	≈ 1.176	Reciprocal minimum (conjectured)	44
Smyth	≈ 1.325	Non-reciprocal minimum (proved)	83

The gap $1.176 < 1.325$ between the reciprocal and non-reciprocal minima reflects a fundamental asymmetry: reciprocal polynomials have roots pairing as $(\alpha, 1/\bar{\alpha})$, which allows more cancellation in the Mahler product.

4 Verification

1. **Jensen:** $M(x^3 - x - 1) = 1.324718$ via roots and via $\exp(\frac{1}{2\pi} \int \log |P| d\theta)$ agree to 10^{-16} .
2. **Exhaustive search:** 56 non-reciprocal monic polynomials of degree 2–4 with coefficients in $\{-1, 0, 1\}$: none has $M < \theta_0$.
3. **Twin:** $M(x^3 - x + 1) = M(x^3 - x - 1) = \theta_0$ (same Mahler measure by symmetry $x \rightarrow -x$).
4. **Comparison:** $M(x^4 - x - 1) = 1.380 > \theta_0$.

5 Pandrosion root structure

For $P = x^3 - x - 1$:

$$F_P(z) = \frac{3z^2 - 1}{z^3 - z - 1} = \frac{1}{z - \theta_0} + \frac{1}{z - \bar{\beta}} + \frac{1}{z - \beta}, \quad (3)$$

where $\beta = -0.662\dots + 0.562\dots i$ with $|\beta| \approx 0.869$. The single pole outside S^1 at $z = \theta_0$ carries all the Mahler mass: $M = \theta_0$ because $\max(1, |\beta|) = 1$.

References

- [1] C. Smyth, *On the product of the conjugates outside the unit circle of an algebraic integer*, Bull. London Math. Soc. **3** (1971), 169–175.
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